

READING PASSAGE 2

Solar Power Generation Technologies

A The study of solar power generation technologies has undergone a remarkable transformation over the past several decades. What was once a niche area of academic inquiry has developed into a rich and diverse field of research, drawing on insights from a wide range of disciplines and theoretical traditions.

B The intellectual lineage of this field extends considerably further than many realise. While modern research techniques have only recently been applied to these questions, philosophers and natural scientists have grappled with related issues for centuries. The transition from speculative philosophical inquiry to systematic empirical investigation represents one of the most significant developments in the intellectual history of this domain.

C At the core of current understanding lies a set of interrelated mechanisms that collectively explain the observed patterns and outcomes. First, structural factors create the underlying conditions within which individual and collective action takes place. These structural dimensions include economic incentives, institutional frameworks, and cultural norms that shape behaviour in ways that are often not immediately apparent to the actors themselves. Second, dynamic feedback processes amplify or dampen initial effects, generating patterns of change that can diverge substantially from what linear models would predict.

D Case studies from diverse geographical and cultural settings provide compelling illustrations of these dynamics in action. In one particularly well-documented example, researchers tracked developments over a fifteen-year period in a setting that offered a natural experiment in the effects of policy change. The introduction of a new regulatory framework produced a cascade of adaptations that rippled through the entire system, generating outcomes that both confirmed theoretical predictions and revealed previously unrecognised complexities.

E Several unresolved debates continue to animate scholarly discussion in this field. Perhaps the most prominent concerns the relative importance of structural versus agentic factors in determining outcomes. While some researchers emphasise the constraining power of institutional and economic structures, others highlight the capacity of individuals and groups to exercise agency and effect change even within seemingly restrictive contexts. This debate has significant practical implications, as it shapes recommendations for policy interventions and institutional reforms.

F The future trajectory of research in this area will likely be shaped by several converging trends. Advances in computational methods and data science are opening up new analytical possibilities, enabling researchers to detect patterns and test hypotheses at scales that were previously unimaginable. Simultaneously, growing recognition of the importance of interdisciplinary collaboration is breaking down traditional barriers between fields and fostering innovative syntheses of diverse theoretical perspectives.

G The evidence reviewed above demonstrates both the substantial achievements of research in this field and the significant work that remains to be done. As new tools and theoretical perspectives become available, and as the phenomena themselves continue to evolve, the research community will need to maintain both rigour in its methods and openness to new ideas. The stakes are high, for the questions addressed by this research bear directly on some of the most important challenges facing humanity in the twenty-first century.

Question 1

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 1 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.

Question 2

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 2 The author believes that all research questions in this field have been answered.

Question 3

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 3 Dynamic _____ processes can amplify or dampen initial effects in complex systems.

Question 4

Choose the correct letter, A, B, C or D.

- 4 Which of the following best describes the author's view on future developments?
- A. pessimism about the future
 - B. cautious optimism with acknowledgment of challenges
 - C. complete certainty about outcomes
 - D. indifference to practical applications

Question 5

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 5 All critics agree that the research findings are universally applicable.

Question 6

Choose the correct letter, A, B, C or D.

- 6 The meta-analyses cited in the passage reveal that:
- A. excessive research funding
 - B. translating academic findings into practical policy
 - C. lack of public interest
 - D. too many international collaborations

Question 7

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 7 Policy-makers have fully implemented all research recommendations.

Question 8

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 8 According to the text, cognitive processes shape the _____ that people make.

Questions 9–10

Choose the correct letter, A, B, C or D.

- 9 The meta-analyses cited in the passage reveal that:
- A. too much data collection
 - B. the generalisability of findings across contexts
 - C. insufficient government funding
 - D. overly complex statistical methods
- 10 Which of the following best describes the author's view on future developments?
- A. excessive funding
 - B. theoretical and methodological limitations
 - C. public interest
 - D. international collaboration

Question 11

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 11 Dynamic _____ processes can amplify or dampen initial effects in complex systems.

Question 12

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 12 The landmark study discussed in the passage involved participants from multiple countries.

Question 13

Choose the correct letter, A, B, C or D.

- 13 According to the passage, what is one of the key factors identified by researchers?

- A. weak effects across studies
- B. moderate to large effect sizes with consistency
- C. contradictory findings
- D. results only in specific populations